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On the Use of Compressive Sensing for the Reconstruction of Anuran Sounds in a Wireless Sensor Network

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***Abstract*—Natural sounds like wind, water, wildlife, and vegetation are considered acoustic sounds and are referred to as the soundscape. Wildlife sounds usually provide enough data to classify and monitor the fauna. In this context, anurans (frogs and toads) have been used by biologists as early indicators of ecological stress in a given environment. Compressive sensing is a promising technique that can be used to reduce the amount of transmitted data in a wireless sensor network, and, thus, minimize the limited resources of sensors nodes. In this work, we collect samples of the anuran audio using compressive sensing, send them towards the sink node, where the samples are reconstructed and identify the correct anuran specie of each call. Our main goal is to evaluate whether compressive sensing is a viable technique to reconstruct properly the anuran calls and identify the correct specie in an environment monitored by a wireless sensor network. We evaluate the proposed methodology by calculating different sound distances between the original and the reconstructed calls. Results show that sampling only 10% of the original data, the sink node can reconstruct the original audio with a high quality.**

I. Introduction

Environmental monitoring is one of the most important sensor network applications that has a great potential to benefit the scientific communities and society as a whole [1], [2], [3]. The monitoring of species can allow us to better understand and preserve a target ecosystem, mainly the endangered species. However, due to the restricted hardware and software resources, there are limits when applying this technology in practice [4].

Natural sounds like wind, water, wildlife, and vegetation are considered acoustic sounds and are referred to as the soundscape. Wildlife sounds usually provide enough data to classify and monitor the fauna. In this context, anurans (frogs and toads) have been used by biologists as early indicators of ecological stress. The reason is that these animals are closely connected to the environment, providing information about aquatic and terrestrial ecosystems [5]. In addition, anuran calls can be used to classify the species within a given site. However, any audio data collected from the environment and wildlife may represent a heavy communication cost for WSNs.

A promising technique that can be used to reduce the amount of transmitted data by sensors, and, thus, minimize the limited resources of sensors nodes is Compressive Sensing (CS) [6]. This technique can be used to reduce the amount of data gathered by means of sampling and, afterwards reconstruct the original signal. For this, the signal must be representable in a sparse base [7]. In WSNs, the number of samples is directly related to the network energy consumption, as well as the amount of packets sent over the network [8].

Previous efforts for automatic animal classification, based on audio streams, have pointed out some features and classifiers for a successful task [9]. The list of features includes Spectral Centroid (S), Signal Bandwidth (B), Zero-crossing Rate (ZC), Hybrid Spectral-Entropy (H) and Mel Fourier Cepstral Coefficient (MFCCs). Classifiers often adopted are k-Nearest Neighbor (kNN) and Support Vector Machine (SVM). These strategies do achieve significant results for different animals, especially when using noise-free audio samples. However, there is neither a standard method, nor a definitive approach for anuran monitoring and sensor networks.

In this work, we collect samples of the anuran audio using compressive sensing, send them towards the sink node, where the samples are reconstructed and identify the correct anuran specie of each call. In this context, our major goal is to evaluate whether compressive sensing is a viable technique to reconstruct properly the anuran calls and identify the correct specie [10]. In this way we expect to reduce the amount of hardware and software resources of sensor nodes, and, thus, saving these precious resources of the WSN. We evaluate the proposed methodology by calculating the different sound distances between the original and the reconstructed calls. The proposed methodology can be applied for different wildlife sound-based monitoring. Anurans are used as a proof-of-concept due to their importance in current ecosystems monitoring.

The remaining of this work is organized as follows. Section 2 presents a brief overview of compressive sensing. Section 3 introduces the methodology proposed in this work to sample, reconstruct, and evaluate the quality of the original call. Section 4 discusses the experimental validation. Finally, Section

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5 presents the concluding remarks and future work.

II. Compressive Sensing

The Nyquist theorem states that a sampling rate of at least twice the maximum frequency present in the signal is necessary for recover it without aliasing nor information loss. By using Compressive Sensing (CS) techniques, we can reconstruct signals with a much smaller number of samples than those required by the Nyquist theorem [10].

Compressive sensing can be applied to reconstruct a signal whenever it has two basic properties: sparsity and linear independence (or incoherence) [7]. Once granted the sparsity and linear independence of the signal, it is necessary to generate a base with independent components of the signal to use it during the reconstruction.

The application of compressive sensing in image processing is very common. Ji and Carin [11] use Bayesian Compressive Sensing (BCP) methods to reconstruct Mondrian works of art. They show a 89% improvement in the reconstruction time, keeping an error of up to 16% in relation to normal reconstruction techniques.

Another example of application of this technique is its use in magnetic resonance equipments. These devices have a high cost of sampling (large number of samples) and the images they generate can be represented in sparse bases [12]. The quality of the CS reconstruction is needed, especially in medical applications [13]. The application of compressive sensing to this kind of application allows the sampling time to be reduced from hours to few seconds.

Ding et al. [14] apply compressive sensing for reconstructing photographs and for image segmentation. For example, a static part (background) of a movable part (foreground), by capturing of consecutive images. Candes and Waking [7] show how to apply the same methodology to audio signals randomly generated using partial sampling, followed by a reconstruction via minimization.

Rana et al. [15] present the design and implementation of an end-to-end noise mapping system, called Ear-Phone, to generate the noise map of an area using participatory urban sensing. The proposed solution uses mobile phones and a central server to perform the signal reconstruction. They address the problem of incomplete samples that are obtained via crowdsourcing by using compressive sensing, focusing on roadside noise pollution. The results show that Ear-Phone has reasonable accuracy, and resource requirements in terms of CPU load and energy consumption.

III. Methodology

A. Gradient Projection for Sparse Reconstruction (GPSR)

In this work, we use a method for sparse reconstruction based on gradient projection (GPSR) [16] to perform the reconstruction of frogs' calls. This method defines the following optimization problem:

$$\min_x \frac{1}{2} \left(|| \mathbf{y} - \mathbf{A}\mathbf{x} ||_2\right)^2 + \tau ||\mathbf{x}||_1, \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{y} \in \mathbb{R}^k$, $\mathbf{A}$ is a $k \times n$ matrix, τ is a positive parameter. $||v||_2$ is the Euclidean norm of v and $||v||_1 = \sum_i |v_i|$ is the ℓ_1 norm, or the Manhattan distance of v. This kind of problem is common in the signal processing area, which it is directly related to the estimation of observations in a system of type:

$$\mathbf{y} = \mathbf{A}x + n. \quad (2)$$

The presence of the ℓ_1 term in Equation 1 implies that the smallest components of x in Equation 2 tend to zero, resulting in sparse solutions, fulfilling the requirements of CS.

The algorithm proposed in [16] implements a solution for minimizing Equation 1 that allows the reconstruction of subsampled signals through an iterative of approximation method. The great advantage of this method is the application of a step for debiasing the approximation, ensuring that any calculation bias is removed. Thereby, it is possible to obtain a solution closer to the ideal solution.

B. Anuran Classification

Due to the intimate relationship with their habitat, the anurans are considered as a metric for evaluation of environmental impacts in a certain region [5]. However, finding and classifying anurans directly in their habitat is not trivial.

The classification of anuran can be done manually [17], by analyzing the recordings and their spectra by specialists, or automatically, using a wireless sensor network as support for the capture and data processing [18]. When performed manually, the quality of the anuran classification is related to the experience of the specialist who performs it. In practice, this task can be slow, prone to failure and is an intrusive process. Although automatic techniques are faster and more accurate than a manual solution, the amount of information used in the classification generates a performance impact on the wireless sensor network.

The classification results using automatic techniques range from 50% in [19], 82.6% in [18] and 99.3% in [9]. The large difference between the results is mainly due to the information used as base for the classification. In particular, Taylor et al. [19] use the intensity of the pixels from the recordings spectrogram as a reference for the classification.

Once the intensity of each pixel and corresponding neighbors are evaluated, we use a C4.5 decision tree to perform the classification. Moreover, that method causes an excessive energy consumption, because it does not use any kind of compression on the audio samples.

To reduce the impact of energy and storage cost, Dang et al. [20] adopt a partial sampling technique. As a result, the amount of data transmitted is less than needed for the signal reconstruction. Thus, energy is saved but the reconstruction process affects the classification success rate.

Studies described in [18], [9] divide the captured audio into segments called syllables. These segments are considered the smallest unit of a frog's call, thus containing the most important information for its classification.

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Huang et al. cite [18] perform the classification based on the k-Nearest Neighbor (kNN) and Support Vector Machine (SVM) techniques, using the syllables as a reference. Vaca-Castaño and Rodriguez [9] use the Mel-Fourier Cepstral Coeficient (MFCCs) to improve the results. MFCCs are known to be widely used in speech recognition [21]. This work is based on the technique proposed in [18] to perform segmentation and the MFCC to improve the classification with kNN, which has better results in terms of classification [9]. However, instead of using the complete recordings as input for the classifier, the amount of information used is reduced by 90%.

C. Audio Capture and Analysis

In our experiments, we use audio samples taken directly from the anuran habitat for four different anuran species. We collected 30 to 50 MB of *WAV* audio files of each specie.

As described before, to reconstruct a signal based on a sparse sampling of an anuran, we need to define a reconstruction base that preserves the main characteristics of the original signal. In the case of anuran classification, the frequencies of the acoustic signals of each specie are enough to make their classification [18], [9].

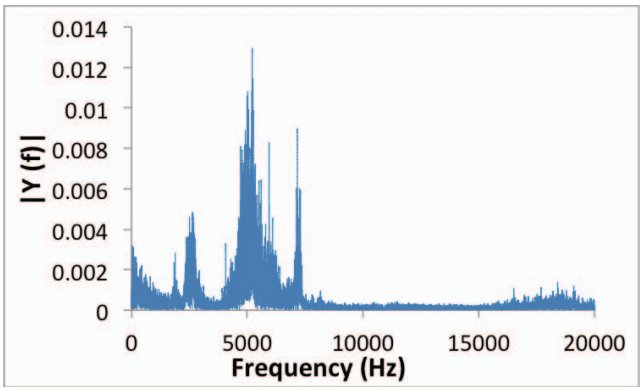


Fig. 3. Frequency rate using Fast Fourier Transform (FFT) on a syllable of the specie *Adenomera andreae*

For the extraction of the characteristics of each monitored species, we performed a segmentation of the audio into smaller units or syllables. Figure 1 shows that anuran calls are composed of peaks, followed by a moment of rest. However, as Figure 2 depicts, despite the main frequencies of a specie have a higher intensity on the peaks, they are still present in moments of rest. In order to maximize the use of the main frequencies, the separation of syllables in this work was performed as shown in Figure 1, where all frequencies are considered to be present between two calls.

Once syllables are segmented, we extract the most representative frequencies and use them as a criterion for reconstruction and classification of the anuran. To separate the frequencies we use a Fast Fourier Transform (FFT) to identify and choose the frequencies of greater occurrence in each specie. The spectrum of a syllable, as well as the main frequencies, are shown in Figure 3.

D. Reconstruction base and experimental parameters

Once the main frequencies of each specie are identified, we create an audio reconstruction base that uses the matrix **A** to evaluate and reconstruct the audio according to a sampled vector x (Equation 2).

For audio data, the reconstruction base must represent the characteristics of a sound wave. Thus, we generated sine waves similar to the ones in the anuran calls. Each frequency was converted into a wave and inserted as a row in matrix **A**, consistent with 10 seconds, which is the size of the samples to be reconstructed.

Once the reconstruction base is built, and by using GPSR [16], original recordings can be reconstructed by using subsamples from an original audio. To accomplish that, we define a downsampling parameter, called α, and generate packets where only one out of every α samples is captured and sent over the network. Complete recordings correspond to a downsampling of $\alpha = 1$, and the compression increases proportionally to α.

The main goal of the compressive sensing techniques is to minimize information loss. Thus, even for signals generated with undersampling, where $\alpha > 1$, the result of the signal reconstruction is equivalent (in number of samples) to a signal where $\alpha = 1$.

IV. Experimental Validation

Once the α parameter is applied on the recorded audio, we can reduce the amount of information in 90% according to Figure 4. After the undersampling of the recordings, the audio files were tested on the classifier kNN and SVM with MFCCs features to verify the quality of the reconstruction. As can be seen in Figure 4, once the compression increases, the quality of the classification decreases. We verified that for $\alpha < 15$, we can maintain the classification rate around 95%, even when using only 10% of the original audio. This is an excellent result for wireless sensor networks.

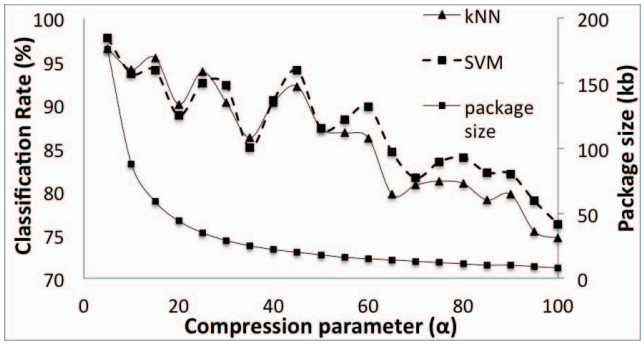


Fig. 4. Audio packet size according to the compression parameter α and the classification rate using kNN and SVM methods

After evaluating the impact on the classification process with the reconstructed audio, it is important to evaluate the consistency of the reconstruction. In this way, we can determine whether the proposed methodology is adequate or

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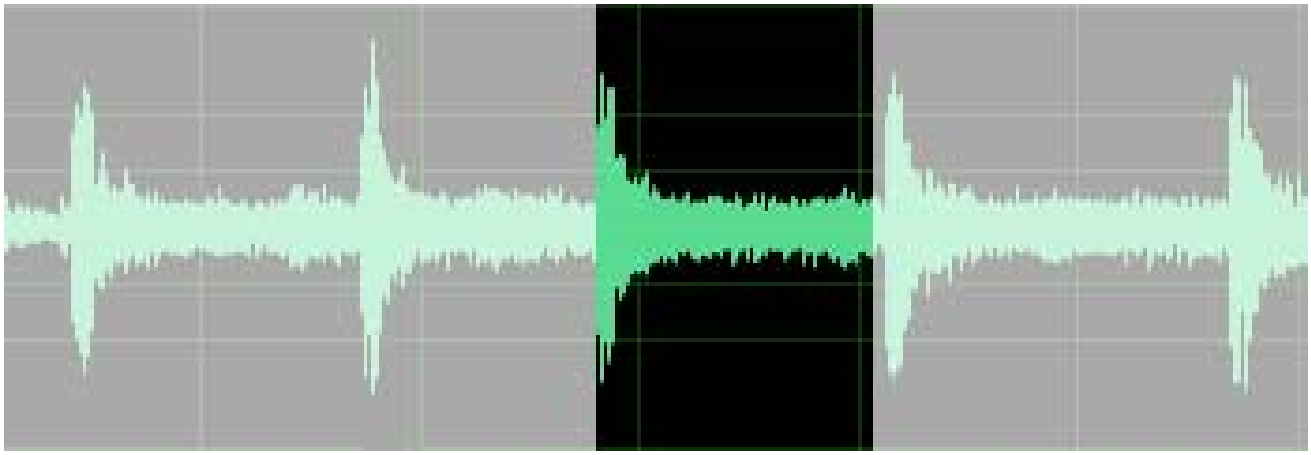


Fig. 1. A call of the specie *Adenomera andreae* with a highlighted syllable

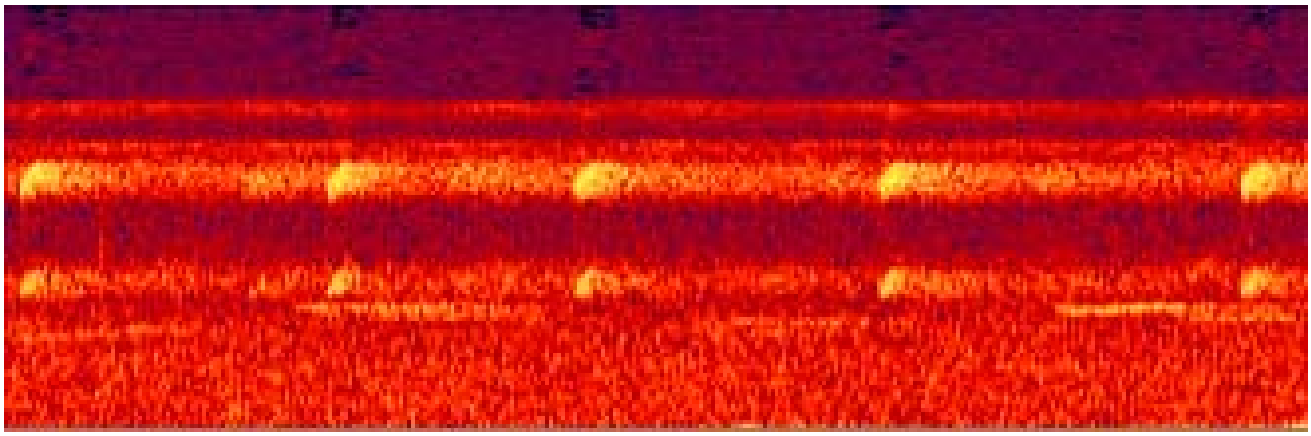


Fig. 2. Spectrogram of a call of the specie *Adenomera andreae*

not for wireless sensor networks. To do that, we calculate four different distance metrics of sound data to compare the reconstructed audio to the original one. All metrics were calculated using the original sample audio and its equivalent reconstruction with an $\alpha = 15$, which represents only 10% of the original data used in the reconstruction.

Initially we calculated the Itakura-Saito distance [22] that is a measure of the perceptual difference between an original spectrum $P(\omega)$ and an approximation $\hat{P}(\omega)$ of that spectrum that can be defined as:

$$D_{IS}(P(\omega), \hat{P}(\omega)) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[\frac{P(\omega)}{\hat{P}(\omega)} - \log \frac{P(\omega)}{\hat{P}(\omega)} - 1 \right] d\omega \tag{3}$$

As we can observe in Figure 5 the distance between the original audio and the reconstructed sample is higher in some points and it tends to zero most of the time. However, if we calculate a squared mean error of all the samples we observe a value of 9%. The error is higher in the points where the syllables are reconstructed because of the slight variation of consecutive syllables. Notice that each specie has

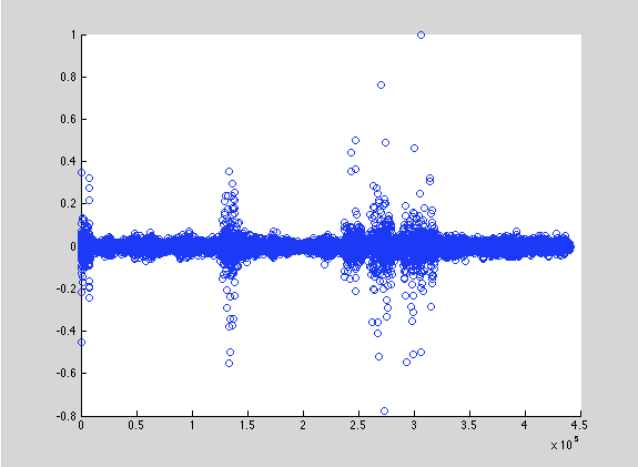


Fig. 5. Itakura-Saito distance of a reconstruction of the specie *Hyla minuta*

a defined set of main frequencies, but it is possible to have minimal variations sometimes. The reconstruction base used

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in this experiments does not consider such variations, causing a minimal error on the Itakura-Saito distance.

The next metric was a Coherence Metric [23] that can be defined as:

$$C_{xy}(\omega) = \frac{|R_{xy}(\omega)|^2}{R_x(\omega)R_y(\omega)}. \quad (4)$$

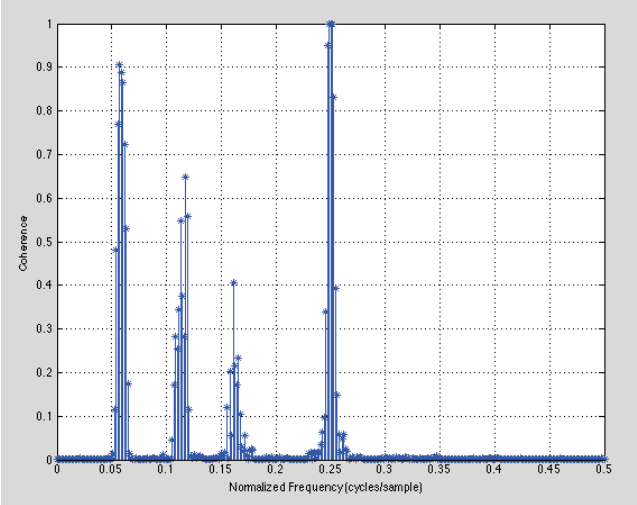


Fig. 6. Coherence measure of a reconstruction of the specie *Adenomera andreae*

The coherence metric is a frequency domain measure, which means that we evaluate the quality of the reconstruction in terms of the used frequencies. As we can observe in Figure 6, there are peaks of coherence when considering the original audio and the corresponding reconstructions. Those peaks of coherence occur on the frequencies used to reconstruct the signal. Considering that the audio of each specie is reconstructed based on the main set of frequencies that define that particular specie, those peaks show a high reconstruction coherence in terms of those main frequencies. Once we reconstruct an audio with high coherence on the main frequencies of the specie we are able to classify them with a high accuracy as we can see in Figure 4.

The third metric iss the Pearson Correlation [23] that can be defined as:

$$\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{E[(X-\mu_X)(Y-\mu_Y)]}{\sigma_X \sigma_Y}. \quad (5)$$

The correlation coefficient ranges from -1 to 1. A value of 1 implies that a linear equation describes the relationship between X and Y perfectly, with all data points lying on a line for which Y increases as X increases. A value of -1 implies that all data points lie on a line for which Y decreases as X increases. A value of 0 implies that there is no linear correlation between the variables.

As we can observe in Figure 7 there is a relation between the Pearson Correlation value and the classification rate. Notice that we have a better classification rate and a higher pearson coefficient as well. It is important to highlight that even if we

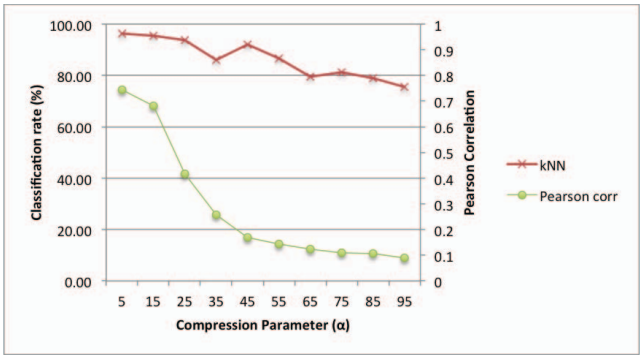


Fig. 7. Pearson correlation behavior according to the compression parameter α

have a nearly perfect classification rate, the Pearson coefficient is 0.74. This occurs because in the original audio there is some noise present, and the frequencies of that noise are not considered on the reconstruction and by ignoring them we decrease the correlation ratio. However, it is possible to preserve a high classification rate even if the correlation is not very good. That is possible due to the nature of the classifier, which only considers the frequencies that are being reconstructed to classify the specie and those frequencies are the main frequencies of the specie.

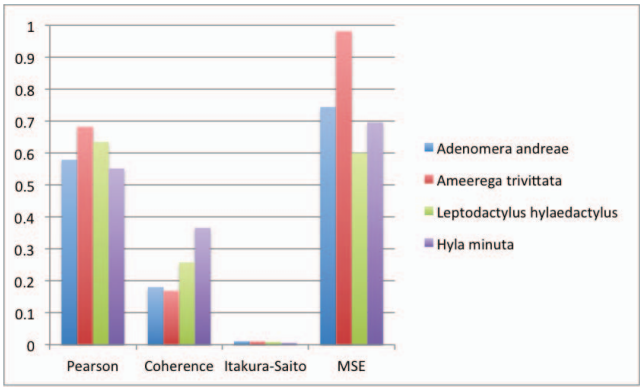


Fig. 8. Metrics behavior among the evaluated species

Finally, we calculate a simple mean squared error between the reconstructed signal and the original one. It is important to highlight that this metric is not an accurate metric to validate the real quality of the reconstruction. By considering the noise on the original audio, the downsampling and the point-by-point difference of the reconstructed signals are expected to be high. However, it is important to compare the behavior of this metric with the other ones, to validate the consistency of the experiments.

As we can see in Figure 8, the metrics behave in a similar way for different sets of species. The Pearson coefficient averages a value of 0.6120 for all evaluated species. In a value between -1 and 1, the results are very promising considering that only 10% of data was used to reconstruct the audio. The

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coherence metric averages 0.2434 for all species. The values of the Itakura-Saito distance were very low, which in the case of this metric is good. By evaluating the distance between both audios we can see that the perceptual difference average is not significant for the reconstruction.

It is expected an improvement on all metrics by using a better scheme to choose the frequencies, which will be present on the reconstruction base. For our experiments, we use an empirical scheme that simply chooses an amount of the most significant frequencies on the FFT calculation (Figure 3).

V. Conclusion and Future Work

This work shows the viability of applying compressive sensing techniques to reconstruct the original audio of anurans with a very good quality. In fact, if we just sample 10% of the original audio, we are able to reconstruct it outside of the wireless sensor network (for instance, at the sink node) with a very high quality. These are tremendous savings in terms of hardware and software resources in a WSN. Furthermore, the distance of the reconstructed signal to the original data is very small. Since each sensor node is only responsible for performing a sampling, it can really save important resources in terms of energy and computing capacity.

Future work includes determining the sampling rate at the sensor node according to the "noisy" conditions in the environment and evaluating the proposed methodology in a real wireless sensor network.

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